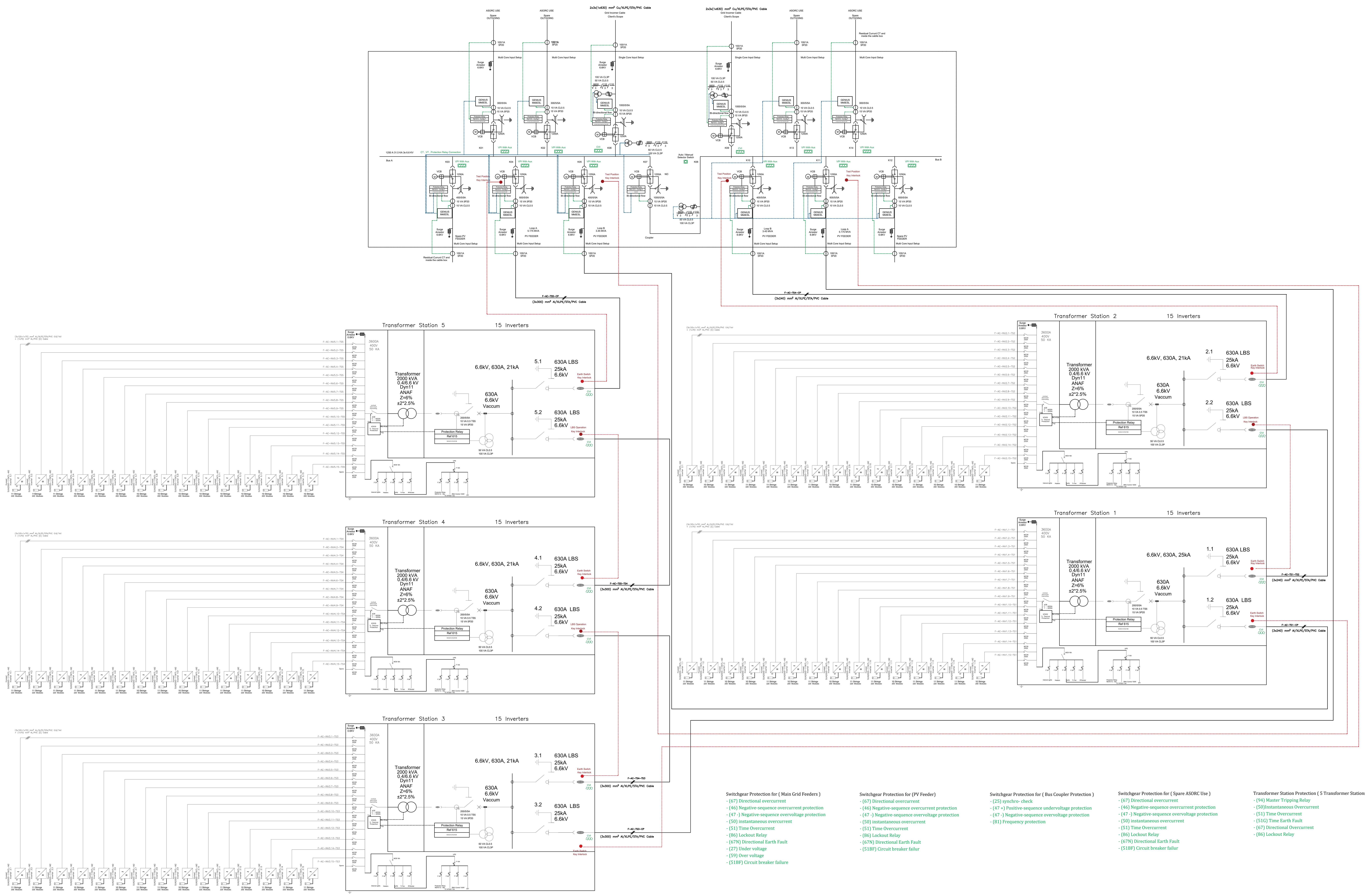


Switchgear Protection for ( Main Grid Feeders )
- (67) Directional overcurrent
- (46) Negative-sequence overcurrent protection
- (47 -) Negative-sequence overvoltage protection
- (50) instantaneous overcurrent
- (51) Time Overcurrent
- (86) Lockout Relay
- (67N) Directional Earth Fault
- (27) Under voltage
- (59) Over voltage
- (51BF) Circuit breaker failure

Switchgear Protection for (PV Feeder)
- (67) Directional overcurrent
- (46) Negative-sequence overcurrent protection
- (47 -) Negative-sequence overvoltage protection
- (50) instantaneous overcurrent
- (51) Time Overcurrent
- (86) Lockout Relay
- (67N) Directional Earth Fault
- (51BF) Circuit breaker failur

Switchgear Protection for ( Bus Coupler Protection )
- (25) synchro- check
- (47 +) Positive-sequence undervoltage protection
- (47 -) Negative-sequence overvoltage protection
- (81) Frequency protection

Switchgear Protection for ( Spare ASORC Use )
- (67) Directional overcurrent
- (46) Negative-sequence overcurrent protection
- (47 -) Negative-sequence overvoltage protection
- (50) instantaneous overcurrent
- (51) Time Overcurrent
- (86) Lockout Relay
- (67N) Directional Earth Fault
- (51BF) Circuit breaker failur

Transformer Station Protection ( 5 Transformer Stations )
- (94) Master Tripping Relay
- (50)Instantaneous Overcurrent
- (51) Time Overcurrent
- (51G) Time Earth Fault
- (67) Directional Overcurrent
- (86) Lockout Relay

| Scenario | Grid Incomer (K01) | ASORC Use 01 (K02) | ASORC Use 02 (K03) | PV Feeder Spare (K04) | PV Feeder Loop A (K05) | PV Feeder Loop B (K06) | Bus Coupler (K07) | PV Feeder Loop B (K09) | PV Feeder Loop A (K010) | PV Feeder Spare (K11) | ASORC Use 03 (K12) | ASORC Use 04 (K13) | Grid Incomer (K14) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Normal Operation | Close | Test Position | Test Position | Test Position | Close | Test Position | Open | Close | Test Position | Test Position | Test Position | Test Position | Close |
| Bus A Outage | Open | Test Position | Test Position | Test Position | Close | Test Position | Close | Close | Test Position | Test Position | Test Position | Test Position | Close |
| Bus B Outage | Close | Test Position | Test Position | Test Position | Close | Test Position | Close | Close | Test Position | Test Position | Test Position | Test Position | Open |
| Night Use | Close | Test Position | Test Position | Test Position | Close | Test Position | Open | Close | Test Position | Test Position | Test Position | Test Position | Close |

## Legend:

- PV Modules
- MV VCB
- RMU MV LBS
- LV Circuit Breaker (ACB,MCB,MCCB)
- Surge Arrestor ( SPD )
- Three Core Cable (3-Phase)
- Merged MFM & Tarrif Meter (MFM & Tarrif Meter)
- VPI With Aux
- Fuse
- Protection Relay
- VT
- CT
- Auto / Manual Selector Switch
- Key Interlocks
- CT Connection
- VT Connection

## Site Info

Orientation: N, E, S, W

Location

| Country | Egypt |
|---|---|
| City | Asyut |

Coordinates

| Point | Northing | Easting |
|---|---|---|
| 1 | 3010670.246 | 301456.303 |
| 2 | 3010867.8521 | 301637.539 |
| 3 | 3011088.120 | 301397.373 |
| 4 | 3011018.446 | 301321.525 |
| 5 | 3011074.216 | 301260.700 |
| 6 | 3010889.898 | 301216.810 |

## General Notes:

## System Specifications

| | |
|---|---|
| No. of Modules | 15,880 |
| No. of Strings | 794 |
| No. of modules per string | 20 |
| Module Wp | 630 |
| No. of Inverters | 75 |
| Inverter Nominal Power | 115 kVA |
| Transformer | 5x2,000 kVA |
| DC Capacity @ STC | 10,004.4 kWp |
| AC Capacity @ 30'c | 8,625 kW AC |
| Substructure | Fixed |
| Tilt | 25° |
| Pitch | 8 m |
| DC/AC | 1.16 |
| String | 1,049.4 Vdc |
| Grid Voltage | 6.6 kVac |

## Drawing Notes:

| Rev. | Description | By | Check | Approved | Date |
|---|---|---|---|---|---|
| 01 | Issued For Approval | RS | AS | AH | 13/11/25 |
| 00 | Issued For Approval | RS | AS | AH | 31/8/25 |

Status: Design Drawing

EPC Contractor:

INFINITY


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Web: www.weareinfinity.com


Project: ASORC, 10MWp Green Electrical Power Generation Plant From PV Solar Energy-On Grid

Client: ASORC-Asyout Petroleum Refinery

Enppi

Drawing Name: AC Single Line Diagram

| Scale (A0) | Drawing No. | Rev. |
|---|---|---|
| NTS | 2358-ASORC-ELEC-001 | 01 |